

Evaluation and Preoperative Management of Kidney Transplant Recipient and Donor

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p0010 Kidney transplantation provides superior long-term outcomes compared with dialysis, in both quantity and quality of life, although the benefit gained varies among individuals.¹ The better outcomes associated with transplantation coupled with the shortage of available organs has led to an expansion of donor criteria and an increasing tendency for centers to accept marginal donor kidneys. This chapter reviews the evaluation and preoperative management of both the kidney transplant recipient and the donor.

s0010 RECIPIENT EVALUATION

p0015 Many transplant centers now accept patients who were previously excluded from transplantation, such as those with human immunodeficiency virus (HIV) infection, obesity, and diabetes. This is because of the availability of newer treatment options for some conditions, a greater understanding of the impact of these conditions on patients and graft survival, and changing societal attitudes regarding equality of access to transplantation. Some absolute contraindications to transplantation remain (Box 107.1), including current infection or malignancy, noncompliance or substance abuse, and any condition likely to severely limit life expectancy (<1–2 years).^{2–4}

p0020 Although the application of guidelines for transplant suitability may be relatively straightforward for patients with a single comorbidity, it is not as simple for those with multiple medical conditions, who represent a growing group of potential transplant recipients. Determination of suitability in such patients often requires input from specialists in a variety of medical and surgical disciplines along with allied health professionals. The final decision needs to be made by clinician and patient together after full discussion of the likely risks and benefits followed by regular reassessment of suitability while the patient awaits transplantation.

p0025 A summary of national and international guidelines^{2–5} is presented in Box 107.2, and selected aspects are discussed.

s0015 Cardiovascular Disease

p0030 Cardiovascular (CV) disease is common in patients with end-stage kidney disease (ESKD) and is a major cause of death in transplant recipients. Hence, CV evaluation is critical in the evaluation of the potential transplant recipient.

s0020 Coronary Heart Disease and Left Ventricular Dysfunction

p0035 The role of pretransplantation investigation and intervention for coronary artery disease (CAD) is controversial. However, given the high incidence of cardiac events in the peri-transplantation period and its major contribution to posttransplantation mortality, we favor aggressive evaluation and intervention in at-risk patients while avoiding unnecessary tests and procedures in low-risk candidates. Patients may be risk-stratified using history and examination, resting

electrocardiography, and chest radiography. Further evaluation is unnecessary in asymptomatic patients without risk factors because of a very low incidence of coronary events.⁶ Further investigation is recommended in patients with abnormal test results or significant risk factors, such as previous cardiac ischemic events, diabetes, smoking, age older than 50 years, hypertension, dialysis duration greater than 2 years, or a family history of CAD.^{2–4}

Symptomatic patients should proceed directly to coronary angiography; noninvasive functional testing should be used to evaluate the need for angiography in asymptomatic patients.^{6,7} Exercise echocardiography or myocardial perfusion scintigraphy are preferred; both testing modalities have negative predictive values for myocardial infarction or cardiac death in excess of 90% in patients with kidney failure.⁸ If significant CAD is identified, treatment before transplantation is required. Treatment consists of aggressive risk factor modification, angioplasty, and stenting or coronary artery bypass grafting in patients with significant stenoses^{6,7} (Fig. 107.1).

In patients with evidence of left ventricular (LV) dysfunction, transthoracic echocardiography should be arranged; a cause should be sought and treated when possible. Severe LV dysfunction may improve significantly after transplantation, so is not an absolute contraindication to transplantation. However, because it is associated with reduced posttransplantation survival, it represents a contraindication in patients with significant comorbidities, unless combined heart and kidney transplantation is appropriate.^{2–4}

Cerebrovascular Disease

Patients with recent transient ischemic attack or stroke are at greatest risk for recurrence early after the primary event; and because stroke after transplantation is associated with a high rate of mortality,⁹ a waiting time of 6 months is recommended. Meanwhile, aggressive risk factor modification is required to limit the likelihood of further stroke. Risk factors for de novo stroke posttransplantation include older age, diabetes, and atrial fibrillation, although transplantation is associated with a reduced risk for stroke relative to remaining on the waiting list or maintenance dialysis. Routine testing for cerebrovascular disease is not advocated in asymptomatic patients. Patients with polycystic kidney disease (PCKD) should be offered investigation for cerebral aneurysm before transplantation, particularly those with additional risks for cerebral aneurysm rupture (e.g., personal or family history of cerebral hemorrhage). Risk for cerebral aneurysm in PCKD is discussed further in Chapter 46.

Peripheral Vascular Disease

Asymptomatic patients with strong femoral and peripheral pulses generally require no further investigation. Patients with diabetes, history of claudication, or reduced pulses require vascular imaging beginning with Doppler ultrasound. Significant disease involving the iliac vessels may make

BOX 107.1 Contraindications to Kidney Transplantation

Current Absolute Contraindications to Transplantation

- Active sepsis
- Current uncontrolled malignancy
- Uncontrolled psychosis
- Active drug dependence
- Any medical condition with a severely shortened life expectancy (<1–2 years)
- Positive T-cell CDC crossmatch

Historical Contraindications to Transplantation^a

- HIV infection
- Hepatitis B and C infection
- Obesity
- Mood disorders
- Age > 60 years
- Previous malignancy
- Blood group incompatibility

^aThese conditions are now acceptable under certain circumstances (see text).

CDC, Complement-dependent cytotoxicity; HIV, human immunodeficiency virus.

BOX 107.2 Recipient Evaluation Checklist

History and Examination

- Cause of kidney failure and risk for recurrence
- Sensitization (transfusion, pregnancy, previous transplant)
- Past and current infections (TB, hepatitis, HIV)
- Immunization (especially hepatitis B)
- Malignancy
- Cardiovascular risks (smoking, hypertension, diabetes)
- Pulmonary, gastrointestinal disease
- Genitourinary tract
- Psychiatric, psychological history
- Surgical issues (weight, iliac vessels, abdomen, previous surgery)

Laboratory and Radiologic Investigations

- Viral serology (HIV, CMV, EBV, hepatitis B and C)
- Liver function tests
- Bone-related issues (PTH, calcium, phosphate)
- Chest radiograph
- Electrocardiogram
- Prostate-specific antigen (for men >50–60 years)
- Mammogram or breast ultrasound (women >50 years or with family history of breast cancer)
- Pap smear (sexually active women)

Immunologic Investigations

- ABO blood group and HLA typing
- Screening for HLA antibodies and autoreactive antibodies
- Crossmatching

CMV, Cytomegalovirus; EBV, Epstein-Barr virus; HIV, human immunodeficiency virus; HLA, human leukocyte antigen; PTH, parathyroid hormone; TB, tuberculosis.

transplantation difficult or impossible and may worsen ischemia in the distal leg. Although it is not an absolute contraindication to transplantation, peripheral vascular disease is associated with increased mortality and should be considered in conjunction with the patient's other comorbidities.

Assessment and Management of Cardiac Status in Potential Transplant Recipients

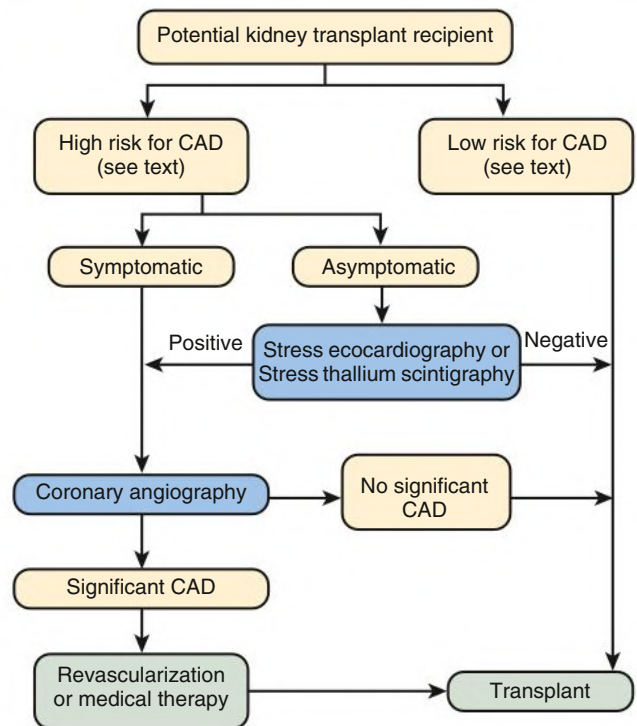


Fig. 107.1 Assessment and management of coronary artery disease (CAD) in potential transplant recipients.

Cancer

Cancer is a major cause of death in kidney transplant recipients. Further increases in the incidence of malignancy are likely with increased graft survival and acceptance of older recipients. The incidence of malignancy is also increased in transplant recipients compared with the general population.¹⁰ However, the effect of transplantation on different types of cancer is not uniform, nor is the effect of different immunosuppressive agents.¹¹ Some cancers, such as nonmelanoma skin cancers (61%–82% at 20 years) and lymphoma, have a markedly increased incidence in transplant recipients compared with the general population, in contrast to breast and prostate adenocarcinoma, which are not as substantially increased.

In patients with previous malignancy, guidelines for recommended waiting time are based on the likelihood of recurrence after transplantation (Box 107.3). In general, the longer the cancer-free interval before transplantation, the smaller the recurrence risk. For most malignancies, a period of 2 to 5 years is recommended.^{2–4} There are several exceptions: longer waiting time (≥5 years) is recommended for breast cancer with nodal involvement, melanoma, and colorectal cancer worse than Duke's stage B1; no waiting time is thought necessary for nonmelanocytic skin cancers confined to the skin, in situ cancers of bladder and cervix, focal microscopic low-grade prostate cancer, monoclonal gammopathy of uncertain significance (without evidence of multiple myeloma), and small (<7 cm) incidentally discovered and surgically removed kidney cell carcinomas. Waiting periods before transplantation should be individualized, taking into account the patient's other comorbidities.

BOX 107.3 Guidelines for Transplantation in Patients With Previous Malignancy

Usual Wait Time of 2 Years

- Most cancers

No Wait Time Necessary

- Asymptomatic T1a (<4 cm) renal cell carcinoma without metastases or suspicious histologic features
- In situ carcinoma
- Focal neoplasm (defined as a localized tumor without metastases)
- Low-grade bladder cancer
- Basal cell skin cancer

Wait Time of >2 Years May Be Necessary

- Melanoma
- Breast cancer
- Colorectal cancer
- Uterine cancer

BOX 107.4 Screening Tests for Occult Infection

Routine Serology

- Cytomegalovirus
- Epstein-Barr virus
- Hepatitis B virus
- Hepatitis C virus
- HIV

Where Indicated, Tests for the Following Conditions and Infections

- HTLV
- Human herpesvirus 8
- Malaria
- Schistosomiasis
- *Strongyloides stercoralis*
- *Trypanosoma cruzi*
- Tuberculosis

Other Routine Investigations

- Chest radiograph
- Urine culture

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Chest radiography is performed as part of the routine assessment. Although extensive screening of all potential recipients is not warranted, potential recipients should be evaluated for breast, cervical, prostate, and colorectal cancer. More comprehensive and targeted evaluation is recommended in patients with a strong family history or suggestive clinical features of malignancy or conditions associated with an increased risk for malignant disease, such as kidney imaging in patients with acquired cystic disease of the kidney for possible kidney cell carcinoma.^{2,3}

s0040 Infectious Complications

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All patients are tested for previous exposure to Epstein-Barr virus (EBV) and cytomegalovirus (CMV) to assess the risk for infection, either primary or reactivation. This guides the appropriate use of prophylactic antiviral agents. For example, patients who are negative for CMV immunoglobulin G (IgG) who receive a kidney from a CMV-positive donor are at the highest risk for infection and may benefit from prolonged prophylaxis compared with the lower risk CMV-negative donor to a CMV-negative recipient (see [Chapter 110](#)). Testing for other infections should be tailored to geographic location; a guide is presented in [Box 107.4](#). All potential recipients should be immunized against hepatitis B virus (HBV). Immunization against encapsulated organisms (pneumococci, *Haemophilus influenzae*, and meningococci) should be considered in patients at high risk for antibody-mediated rejection in case-rescue therapy (e.g., splenectomy) or when complete inhibition with eculizumab is required.

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Centers experienced in managing HIV infection and transplantation report excellent patient and graft survival in carefully selected recipients.¹² In the absence of an acquired immunodeficiency syndrome (AIDS)-defining illness, patients with sustained CD4 counts greater than 200/mL and undetectable HIV viral loads can be considered for transplantation.² Patients with HBV infection may be considered for kidney transplantation if there is no evidence of active viral replication (HBV DNA or HBV early antigen [HBeAg] negative), advanced liver disease or cirrhosis (as determined by liver biopsy), or hepatocellular carcinoma (HCC).²⁻⁴ Immunosuppression can increase HBV replication; hence, treatment before transplantation is indicated in patients with active disease, and although data to support prophylactic antiviral therapy after transplantation are scarce, treatments such as entecavir are commonly used while immunosuppression is at its highest (first 12–24 months). Early reports suggested that mortality may be increased

HIV, Human immunodeficiency virus; HTLV, human T-lymphotropic virus.

in HBV-positive patients after transplantation compared with HBV-negative recipients,¹³ but it is unclear whether this is true in the current era. We recommend frank disclosure of potential risks.²⁻⁴

Patients with hepatitis C should be assessed by measurement of hepatitis C virus (HCV) viral load and a liver biopsy. In patients without cirrhosis, transplantation should not be delayed by treatment because direct-acting antivirals can be safely used to treat HCV after transplantation,¹⁴ and mortality in HCV-positive patients is reduced by transplantation compared with remaining on dialysis.¹⁵ With informed consent, HCV-positive patients may receive a kidney from an HCV-positive donor because any possible increased risk for the latter is offset by a significantly reduced waiting time. Patients with HBV and HCV infection should be screened every 12 months for HCC by liver ultrasound and serum α -fetoprotein. Those with cirrhosis may be considered for combined kidney-liver transplantation.

Patients at high risk for tuberculosis (TB) reactivation after transplantation (previous TB, abnormal chest radiograph, or positive tuberculin skin test result; residence in an endemic area) who have not been previously treated should receive prophylactic isoniazid after transplantation (see [Chapter 110](#)). Interferon- γ (IFN- γ) release assays may replace or supplement the tuberculin skin test because they appear more sensitive for detecting latent TB infection.¹⁶ The need to evaluate patients for TB is determined by the likelihood of previous exposure.

Previous graft loss caused by polyoma (BK) viral nephropathy is not a contraindication to repeated transplantation. Waiting for serum and urine BK polymerase chain reaction test results to become negative appears to be preferable,¹⁷ whereas the value of graft nephrectomy before repeated transplantation is unclear. Vigilant evaluation for recurrence is recommended.⁴

Obesity

Transplantation in obese patients (body mass index [BMI] > 30 kg/m²) generally improves survival compared with matched waiting list controls,¹⁸ but inferior outcomes for patient and graft survival, delayed graft function, wound healing, and infectious complications have been reported compared with nonobese patients, particularly those with BMI

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greater than 35 kg/m².^{19,20} Additionally, overweight and obese patients are more likely to develop diabetes after transplantation, which can adversely affect graft and patient survival. Although not an absolute contraindication to transplantation, patients with obesity should be offered weight loss strategies (including bariatric surgery where appropriate) before transplantation to reduce their risk of mortality and morbidity.

s0050 **Recurrent Disease**

p0105 The risk for disease recurrence should be discussed as part of the informed consent process, particularly in primary kidney diseases with a high risk for recurrence (e.g., focal segmental glomerulosclerosis).²¹ Recurrent disease accounts for more than 5% of graft loss²¹ and is an increasingly important cause of graft dysfunction with increasing time posttransplantation. The risks and management of recurrent disease are discussed in [Chapter 113](#).

s0055 **Gastrointestinal Disease**

p0110 Investigations searching for gastrointestinal disease are not warranted in the asymptomatic patient.^{3,4} Patients with active acute or chronic pancreatitis should not undergo transplantation until they have been clear of symptoms for at least 3 months. Patients with active peptic ulcer disease should be treated before transplantation with proton pump inhibitors, and this should be continued to prevent ulceration after transplantation. Patients with symptomatic diverticular disease require colonoscopy and potential colonic resection in severe cases before transplantation because they are at increased risk for perforation on immunosuppressive medications.³ Although symptomatic cholecystitis should be treated surgically before transplantation, asymptomatic cholelithiasis does not require surgery before transplantation because cholecystectomy after transplantation is required in less than 10%, and results in no increased mortality or morbidity compared with pretransplantation cholecystectomy.²²

s0060 **Genitourinary Disorders**

p0115 Testing for genitourinary tract disorders before transplantation is indicated in those with features suggestive of urinary obstruction, especially in children, in whom urologic problems are a major cause of ESKD. If obstruction is found, urologic assessment and/or voiding cystourethrography and urodynamic studies are indicated. Depending on the findings, intervention may be warranted to ensure bladder emptying and limit bladder pressures after transplantation; this may involve bladder augmentation, urinary diversion, or self-catheterization.

p0120 Native nephrectomy before transplantation should be considered in patients with recurrent or persistent kidney sepsis, particularly in the setting of nephrolithiasis. Very large polycystic kidneys may need to be removed to accommodate the transplant kidney. Whether previous grafts should be removed before repeated transplantation is controversial. Nephrectomy of a failed graft is commonly performed on withdrawal of immunosuppression in patients with early graft failure (<12 months) to alleviate symptoms such as pain, fever, and weight loss.²³ Other indications include graft sepsis and allowance of room for the new graft. However, unless there is a convincing reason to remove the graft, it is generally left in situ. In these circumstances the patient may need to stay on minimal immunosuppression, such as prednisolone for 3 to 6 months, to minimize graft tenderness and inflammation. Graft nephrectomy may be associated with an increased risk for human leukocyte antigen (HLA) sensitization,²³ but this is not a universal finding. Another advantage of leaving the previous transplant in situ is preservation of any residual kidney function and urine output.

s0065 **Pulmonary Disease**

p0125 Physical examination and chest radiography are indicated for all potential recipients; pulmonary function tests or computed tomographic (CT) scanning are done if clinically indicated. Guidelines suggest that patients

with a short life expectancy associated with pulmonary disease, such as cor pulmonale, uncontrolled asthma, and severe obstructive lung disease (FEV₁ < 25% of predicted or Po₂ < 60 mm Hg [8 kPa] on room air), or those needing home oxygen should be excluded from transplantation.⁴ Many centers require patients to cease smoking before acceptance because smokers have an increased risk for death and graft loss. Cessation of smoking demonstrates positive lifestyle behavior and good adherence, suggesting these factors will remain optimal after transplantation.

Psychosocial Issues

Psychosocial issues can have a major impact on transplant outcomes. Patients should be evaluated to judge capacity to consent and assess likely adherence to the transplant medication regimen. Predicting adherence can be challenging and may be based on pretransplantation adherence, such as to dialysis management regimens; transplantation should not proceed if the multidisciplinary assessment suggests that adherence is unlikely.²⁻⁴

Cognitive impairment is not an absolute contraindication to transplantation if appropriate supports and proxy arrangements are in place. Patients with serious psychiatric illnesses require assessment by a psychiatrist to determine transplant suitability and devise a management plan to cope with possible consequences of immunosuppressive medications such as corticosteroids.²⁻⁴ Drug and alcohol addiction should be addressed with rehabilitation and demonstrated abstinence before the patient is listed for transplantation.

Presence of Multiple Comorbidities

Increasingly, patients with multiple comorbidities are referred for consideration of transplantation. Each morbidity in itself often is not an absolute contraindication but taken together they may significantly affect prognosis. Controversially, some have suggested that patients must have specified long-term survival prospects—for example, an 80% chance of surviving 5 years—to be accepted for transplantation from a deceased donor. In addition, risk calculators have been devised based on registry data to allow such predictions (e.g., <https://optn.transplant.hrsa.gov/resources/allocation-calculators/epts-calculator/>).²⁴

Reevaluation of Patients on the Waiting List

Patients may wait several years on the waiting list before an opportunity arises to undergo transplantation. It is vital that when their opportunity comes, they are still suitable candidates. Waiting list patients therefore should be reassessed at regular intervals ([Box 107.5](#)). General measures, such as cancer screening (e.g., skin, prostate, breast, and cervical), should be continued as indicated. Reassessment of cardiac status is advocated based on risk. Diabetic and other high-risk patients should be reassessed every 1 to 2 years.⁶ The value of reassessing low-risk patients is more questionable, but given that CKD is a strong risk factor for cardiac disease, repeated stress testing by exercise or pharmacologically driven echocardiography or myocardial perfusion scintigraphy at least every 3 years seems appropriate. Patients with preexisting medical conditions (e.g., HIV infection or viral hepatitis) require regular specialist reviews, with any issues arising brought to the attention of the transplant team. Surgical reassessment may be needed in patients with peripheral vascular disease, if patients gain weight, or if a complication such as peritonitis occurs while the patient awaits transplantation. Patients should be temporarily removed from the waiting list if they develop a serious infection or other illness until it is resolved.

DONOR EVALUATION

Classification of the Deceased Donor

Deceased donors can be classified as either heart-beating donors with loss of brainstem function (donation after brain death [DBD]) or non-heart beating donors (donation after cardiac death [DCD]). In the last

BOX 107.5 Potential Recipient Reevaluation While Awaiting Transplantation

- Stress echocardiogram or thallium scan
- Angiography if indicated (see Fig. 107.1)

Cancer Surveillance Relevant to Age, Sex, and Risk Factors

- Prostate-specific antigen
- Mammography
- Pap smear
- Colonoscopy if indicated
- Skin cancer check

Comorbidity Reassessment

- Viral hepatitis
- Liver function tests
- α -Fetoprotein
- Liver ultrasound
- HIV infection
- CD4 count
- AIDS-defining illness
- Other (see discussion of individual organ systems)
- Cardiac reevaluation every 1 to 2 years depending on risk factors (e.g., diabetes)

two decades the proportion of DCD donors has been increasing in many countries as a result of policies aiming to maximize the opportunities for donation.²⁵

In the United States, DBD donors were previously divided into standard criteria donors (SCDs) and expanded-criteria donors (ECDs) for allocation purposes. ECD referred to heart-beating donors aged older than 60 years or 50 to 59 years with two or three of the following criteria: a history of hypertension, elevated serum creatinine at donation (>1.5 mg/dL or 130 μ mol/L), or death from a cerebrovascular accident. In some jurisdictions, the term *marginal donor* is used to describe donors with significant underlying disease (hypertension, diabetes, vascular disease, kidney impairment) or advanced age (>65 years). In the last decade the United Network for Organ Sharing (UNOS) in the United States adopted a new system for classifying and allocating kidneys using a quality index.²⁶ Other countries have also begun to use a similar system.²⁷ The Kidney Donor Risk Index (KDRI) is a score that combines various donor factors to summarize the perceived quality of deceased donor kidneys. Factors in the calculation include age, creatinine, mode of death, donor pathway (brain death vs. circulatory death), history of hypertension or diabetes, race, and risk for HCV infection. A raw index score is converted to a percentile producing the Kidney Donor Profile Index (KDPI), rating from 0 (best) to 100 (worst).

The Maastricht classification defines DCD donors as controlled or uncontrolled (Box 107.6).²⁸ Controlled donors are those who experience cardiac arrest after withdrawal of support or after brain death. Uncontrolled donors are those who are deceased on arrival to the hospital or who had unsuccessful cardiopulmonary resuscitation. From a practical point of view, a system that uses controlled donors is easier to implement than one using uncontrolled donors. This is largely related to factors surrounding ethical considerations and the consent process involving relatives of the donor.

The survival of kidneys from ECD donors (loosely correlating with a KDPI $> 85\%$) and from some categories of DCD donors is generally inferior to that of kidneys retrieved from SCD donors.²⁸ Many matching schemes attempt to allocate these less ideal grafts to recipients who are predicted to have a lower than average overall

BOX 107.6 Classification of Donation After Cardiac Death Donors^a

Uncontrolled

Category I: Dead on Arrival to Hospital

Cause of death is usually obvious (e.g., severe head injury), and no resuscitation is given.

Category II: Unsuccessful Resuscitation

The patient is brought to the emergency department while being resuscitated, but this is not effective. Alternatively, cardiac arrest occurs in hospital and the patient cannot be resuscitated.

Controlled

Category III: Awaiting Cardiac Arrest

Severe brain injury without brain death. Patients are usually ventilator dependent. Cardiac arrest occurs once support is withdrawn.

Category IV: Cardiac Arrest While Brain Dead

Patient has a cardiac arrest after being declared brain dead. Alternatively, this occurs during brain death testing and the patient is not successfully resuscitated.

Category V: Unexpected Cardiac Arrest in a Critically Ill Patient

Example: unsuccessful resuscitation after unexpected cardiac arrest in intensive care.

^aAs per Maastricht classification²⁴ (also known as non-heart-beating donors).

From Indudhara R, Kenney PJ, Bueschen AJ, Burns JR. Live donor nephrectomy in patients with fibromuscular dysplasia of the renal arteries. *J Urol*. 1999;162:678–681.

survival, but practice varies considerably across countries.²⁹ Simple age matching can be a part of this allocation process, or, alternatively, a more complex longevity-matching process can be performed using a recipient index.³⁰ The Expected Post Transplant Survival (EPTS) score is the best known example, which incorporates factors such as age, diabetes, years on dialysis, and previous transplantation into the score.³¹

An attempt to better use kidneys from donors perceived to have an increased risk of viral infection (HBV, HCV, and HIV) despite negative nucleic acid testing (NAT) has been made in some countries in recent years.³² The issue here consists of a residual risk of viral transmission, related to a window period of possible infection in the donor, leaving the viral tests negative despite the presence of infection. In reality, the risk is low provided that results of NAT testing are negative. Additionally, HCV can now be effectively treated, leading to some units actually transplanting organs known to be infected and then treating the recipient for HCV.³³

Evaluation of the Deceased Donor

In most circumstances, organ donor coordinators screen potential deceased donors after referral from intensive care units (ICUs) or emergency departments (EDs). Patient records are assessed and relatives are interviewed about important aspects of the clinical history. The assessment focuses on general health (including history of infections and cancer), social history (especially drug use and sexual history), and laboratory evidence of kidney impairment or other diseases (Box 107.7). Patients with sepsis, acute hepatitis, or HIV infection are excluded from donation, as are those with a history of malignancy. Nonmelanoma skin cancers do not lead to exclusion, nor do primary brain tumors unless they are of a high grade or the donor has received

BOX 107.7 Deceased Donor Evaluation Checklist**Medical History**

- Hypertension, diabetes
- Malignancy
- Infections: Past and current (TB, hepatitis, HIV)
- Transfusions
- Trauma
- Surgical history
- Hospitalizations

Social History

- Intravenous drug use
- Alcohol, smoking
- Sexual behavior
- Tattoos, acupuncture
- Overseas travel
- Incarceration

Examination

- Blood pressure
- Cardiac, vascular
- Lymphadenopathy
- Abdominal

Laboratory and Technical

- Serum creatinine
- Urinalysis, urine culture
- Liver function tests
- Coagulation profile, complete blood count
- Blood culture
- Virology,^a depending on geographic region: Antibodies to CMV, EBV, HSV-1 and HSV-2; HHV-6, -7, or -8; HCV, HBV (including HBsAg, anti-HBcAg, IgG, and IgM); HIV; West Nile virus; rabies; HTLV-1
- Parasites, depending on geographic region: malaria, babesiosis, toxoplasmosis, Chagas disease, syphilis
- Fungi in appropriate regions: *Coccidioides*, *Histoplasma*
- Tuberculosis (depending on geographic region)
- Chest radiograph
- Electrocardiogram
- Kidney biopsy if there is concern for chronic kidney disease

Operating Room Evaluation

- Intraabdominal examination to detect occult malignancy
- Macroscopic appearance of kidneys

^aChoice of virologic investigations according to local risks.

CMV, Cytomegalovirus; EBV, Epstein-Barr virus; HBcAg, hepatitis B core antigen; HBsAg, hepatitis B surface antigen; HBV, hepatitis B virus; HCV, hepatitis C virus; HHV, human herpesvirus; HIV, human immunodeficiency virus; HSV, herpes simplex virus; HTLV, human T cell lymphotropic virus; TB, tuberculosis.

chemotherapy or had a craniotomy or cerebral shunt inserted.³⁴ In some centers, donors potentially carrying HBV or HCV are accepted only for recipients who are seropositive for these viruses. The risk for an unknown donor malignant neoplasm is approximately 1.3%; however, the risk for transmitting a donor malignant neoplasm is lower at approximately 0.2%.³⁵

Evaluation of kidney function is determined by history, urinalysis, and serum creatinine concentration. In some patients a biopsy (often performed at retrieval) may provide useful information, particularly with ECDs or donors with a high KDPI. Estimated glomerular filtration

rate (eGFR) at admission should not be overtly reduced (>60 mL/min/1.73 m²), but a temporary decline in kidney function is acceptable if function is expected to recover. Proteinuria (>0.5 g/day) may indicate structural kidney damage and is a valid reason for nonacceptance.

The use of kidneys from very young donors varies among centers. Donors aged younger than 5 or 6 years are associated with high risk for graft failure, especially from vascular thrombosis.³⁶ For this reason, some centers occasionally transplant two kidneys in the one recipient en bloc, using the aorta and inferior vena cava as conduits.³⁷

Deceased Donor Management Before Transplantation

In the DBD donor, maintenance of adequate blood pressure (BP) and oxygenation are important to prevent warm ischemic kidney injury. The use of pressor agents, volume resuscitation, and other conditioning strategies is complex and has been the subject of several guideline documents (see the Intensive Care Society website, www.ics.ac.uk). In this category of donor, the kidneys are generally not subject to significant warm ischemia at the time of organ retrieval unless the donor experiences prolonged hemodynamic compromise.

In DCD, once death is certified and deemed irreversible, either rapid surgical exposure of the great vessels with cooling of the organs followed by prompt retrieval is required, or, alternatively, the kidneys are cooled in situ by insertion of perfusion catheters through the femoral vessels (see also Chapter 108). Surgical retrieval can then take place after the following occur, as required: family counseling, donor assessment, or relocation from one hospital area to another (e.g., from the ICU or ED to the operating room). DCD is inevitably associated with warm ischemic kidney injury. This is responsible for the higher rate of delayed graft function that is seen in this group. The need for dialysis after transplantation is approximately 50% but varies from 30% to 90%, depending on the Maastricht category of donor.²⁸

Living Donors

Live kidney donation is currently accepted in most countries based on the demand for deceased donor organs—which far outweighs the supply—and the apparent very low level of risk to the majority of healthy donors.³⁸ Added to this are the detrimental effects for the recipient of waiting on dialysis and the generally superior results obtained through use of living donors.

Living donors may be related, unrelated, altruistic, or part of a donor exchange or list-exchange program. In many countries with well-established transplant programs, half or more of all transplants are now performed with living donors. In Japan, Brazil, and the Middle East, more than 80% of transplants use living donors. The superior outcomes of transplantation from living donors compared with that from deceased donors has supported the development of living donor paired exchange, in which living donors who are incompatible with their intended recipients are exchanged between recipients. Another approach is living donor–deceased donor exchange, in which the donor donates to the wait list in exchange for the intended recipient receiving priority on the list.³⁹

In some countries, either a state-organized or a free-market system results in the purchase of living donor kidneys.³⁹ The Declaration of Istanbul and the World Health Organization both condemn the exploitation of living donors who are vulnerable (illiterate, impoverished, undocumented immigrants, prisoners, and political or economic refugees).⁴⁰

Many centers have extended their selection of donors to include patients who are mildly hypertensive, overweight, or hyperlipidemic or who have other abnormalities (such as isolated microhematuria or

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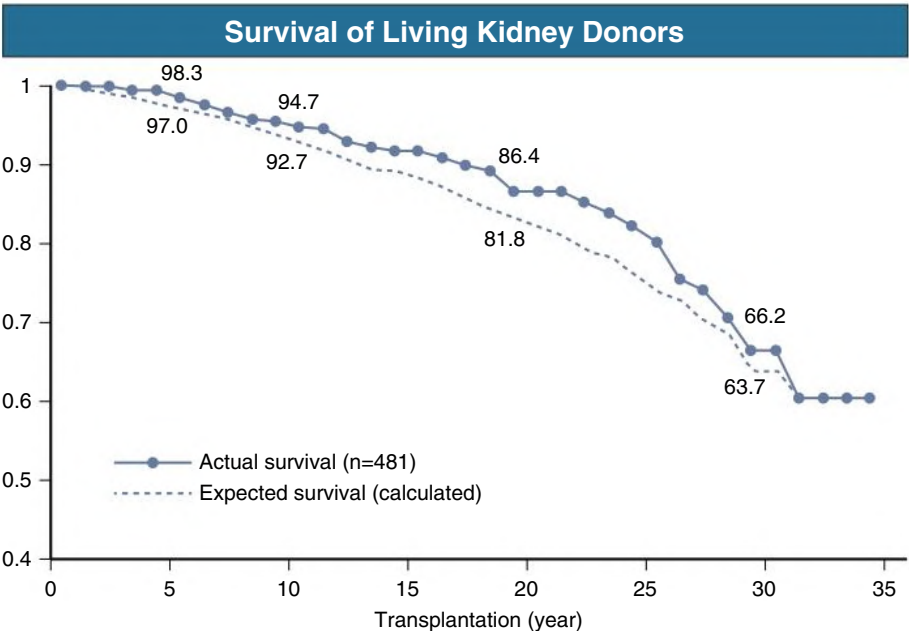


Fig. 107.2 Survival of Living Kidney Donors. Donor survival rates after living-donor kidney donation. Retrieval rate was 80.0% (481/601). Expected survival was calculated using data for an age- and sex-matched normal population in Japan provided by the National Cancer Center. (From Okamoto M, Akioka K, Nobori S, et al. Short- and long-term donor outcomes after kidney donation: analysis of 601 cases over a 35-year period at Japanese single center. *Transplantation*. 2009;87[3]:419-423.)

previous nephrolithiasis).⁴¹ Although donation appears to be safe in the short- to medium-term for most of these patients, long-term risks have not been adequately assessed.

Mortality and Morbidity

Mortality related to living donation is a catastrophic and unexpected event; the perioperative risk for donor death is approximately 3 in 10,000.^{4,42} In the longer term, survival of donors appears to be similar to that of the general population (Fig. 107.2).

Physical and psychological function in living donors is higher than the community norm. Physical issues reported by donors after donation frequently include a temporary decrease from baseline in energy; some note a longer time to full recovery than anticipated and incision pain (after open nephrectomy) that lasts longer than expected. Psychological factors usually include an improved relationship with the recipient, an improved self-image, and frequently a positive effect on the donor's life. Longer-term psychological morbidity appears minimal; however, some series have reported an association with anxiety, depression, or other psychological issues in a small proportion of the patients.⁴³ Although most donors have a positive experience, a small number for a variety of reasons regret the decision to donate (0%–5%).⁴⁴ Psychological evaluation before donation is therefore extremely important, as is support and counseling after donation, especially when the transplant does not go as well as anticipated.

Donor Evaluation: Living Donors

Several groups have developed guidelines for the evaluation and aftercare of living donors.^{45–47} An outline of the usual donor evaluation is shown in Boxes 107.8 and 107.9. An assessment of relevant anatomy may be achieved by CT angiography or magnetic resonance angiography, depending on the center. Kidney arteriography is not usually necessary, given the anatomic detail available from noninvasive techniques.

BOX 107.8 Living Donor Evaluation Checklist: History and Examination

History

- Hypertension
- Diabetes (including gestational)
- Infections
- Cancer (including skin lesions)
- Vascular disease
- Kidney calculi
- Gout
- Urinary tract
- Family history
- Medication use (including NSAIDs, herbs)
- Smoking
- Illicit and intravenous drug use
- Sexual history
- Vocation, sport interests
- Level of physical activity, exercise
- Psychiatric history, psychological factors
- Willingness to donate
- Relationship with recipient

Examination

- Blood pressure
- Weight and height, BMI
- Joints, skin
- Cancer (including skin lesions, breast)
- Lymph nodes
- Vascular disease
- Heart and lungs
- Abdomen

BMI, Body mass index; NSAIDs, nonsteroidal antiinflammatory drugs.

BOX 107.9 Living Donor Evaluation Checklist: Investigations

Laboratory and Radiologic Investigations

- Urinalysis (blood, protein)
- Urine microscopy and culture (blood, organisms)
- Serum electrolytes, urea, and creatinine
- Liver function tests
- Full blood examination
- Fasting blood glucose and/or oral glucose tolerance test
- Fasting lipids
- 24-Hour urine, creatinine clearance, or GFR measurement by iohalamate, Cr-EDTA, DTPA clearance, 24-hour urine protein, or protein excretion by other methods (e.g., protein-creatinine ratio)
- Serum uric acid, calcium, phosphate
- Viral screening: HBV, HCV, HIV, CMV, EBV serology
- Syphilis screening (RPR)
- TB screening (PPD)
- Electrocardiogram
- Chest radiograph
- Females: Pap smear, mammography (according to age and family history)
- Males: Prostate-specific antigen (according to age and family history)
- Additional cardiac investigations (where indicated by age, history, risk factors)
 - Stress test
 - Echocardiography
 - Ambulatory blood pressure

Kidney Imaging (According to Local Expertise)

- Computed tomographic angiography
- Magnetic resonance imaging angiography
- Catheter angiography

CMV, Cytomegalovirus; Cr-EDTA, chromium-labeled ethylenediaminetetraacetic acid; DTPA, diethylenetriaminepentaacetic acid; EBV, Epstein-Barr virus; GFR, glomerular filtration rate; HBV, hepatitis B virus; HCV, hepatitis C virus; HIV, human immunodeficiency virus; PPD, purified protein derivative test; RPR, rapid plasmin reagent; TB, tuberculosis.

Assessment of Kidney Function

Most centers use GFR of 80 mL/min/1.73 m² as the lower limit for donors. It is accepted that this threshold may be too low for donors younger than 40 years and too high for donors older than 60 to 65 years. An alternative approach is to consider the age-specific GFR and accept donors only if they fall within the average for this age range. This method has been recommended by the British Transplantation Society (guidelines available at www.bts.org.uk) and is presented in Fig. 107.3. An alternative approach uses the life expectancy of the donor.⁴⁸ Based on these calculations, a 30-year-old donor would require a GFR of 123 mL/min/1.73 m²; the required GFR for a 70-year-old person would be 68 mL/min/1.73 m².

Hypertension and Proteinuria in the Living Donor

Donation may be acceptable for some hypertensive individuals if BP is well controlled, GFR is as expected for donation and age, and there are no features of end-organ involvement from hypertension.⁴⁵⁻⁴⁷ The evaluation for hypertension should include BP measurements on three separate occasions. Borderline elevated levels should be further evaluated with ambulatory BP monitoring. If elevated BP is detected and the prospective donor is still under consideration, echocardiography (looking for LV hypertrophy), ophthalmologic evaluation (looking for hypertensive retinal changes), and assessment for albuminuria

Acceptable GFR in Living Donors by Age

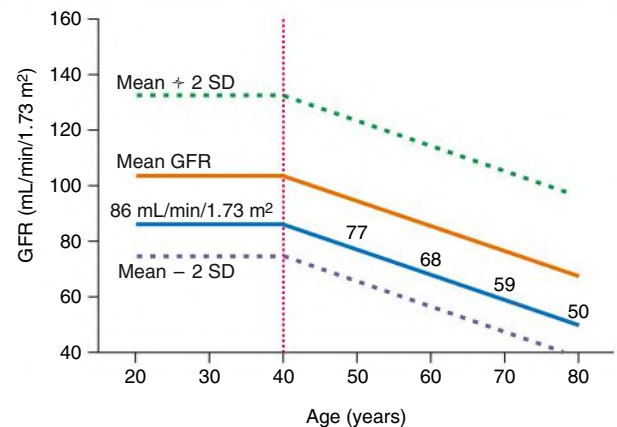


Fig. 107.3 Acceptable glomerular filtration rate (GFR) in living donors by age. Diagram explaining the minimum acceptable age-associated GFR in living donor candidates. The solid orange line shows the variation with age of mean GFR. The outer dashed lines show the +2 and -2 population standard deviation (SD) limits. GFR is constant up to the age of 40 years and then declines at the rate of 9 mL/min/1.73 m² per decade. The reference plot is based on an analysis of data for 428 living kidney transplant donors who had chromium-51-labeled ethylenediaminetetraacetic acid (⁵¹Cr-EDTA) GFR measurements. The solid blue line shows the acceptable lowest GFR for young adults, declining to 50 mL/min/1.73 m² at age 80 years. For transplant donors with preoperative GFR values above the solid blue line, the GFR of the remaining kidney will still be above 37.5 mL/min/1.73 m² at age 80 years. (Modified from the revised British Transplantation Society/Kidney Association U.K. guidelines for living donor kidney transplantation. www.bts.org.uk/.)

(suggesting hypertensive kidney damage) should be undertaken. The prospective donor should be excluded if any of these features are present.

KDIGO (Kidney Disease: Improving Global Outcomes) recommends using the ratio of albumin to creatinine from a random urine sample with confirmation using an albumin excretion rate (AER) to assess donors for significant albuminuria (www.kdigo.org/guidelines/).^{47,49} An acceptable AER is less than 30 mg/day, with 30 to 100 mg/day requiring an individualized approach. An AER greater than 100 mg/day should exclude potential donors.

Obesity and Abnormal Glucose Tolerance in the Living Donor

Although many centers accept obese living donors, several issues must be addressed. These include the impact of obesity on perioperative complications, future kidney function, and CV health. In one study, obese (BMI > 30) patients had an increased rate of proteinuria and kidney impairment 10 to 20 years after nephrectomy.⁵⁰ Obese individuals may therefore be more prone to development of kidney disease after donation, but this issue has not been carefully studied.

Future risk for diabetes is another important consideration. In addition to close assessment of those who are overweight, prospective donors with an abnormal fasting glucose concentration, a history of gestational diabetes, or a first-degree relative with diabetes should be evaluated with an oral glucose tolerance test. An abnormal glucose tolerance test result is a contraindication to donation. Patients often lose weight and otherwise change their lifestyle (exercise, diet), leading to an improvement in their results and eventual acceptance as donors. It is important that these lifestyle and risk modifications be sustained after donation occurs.

s0135 **Kidney Abnormalities in the Living Donor**

p0260 As well as factors identified in the history (e.g., previous calculi, urinary tract infections, prostatic disease), a variety of previously unidentified kidney abnormalities can be encountered in prospective donors during their assessment. These include microhematuria, kidney scarring (e.g., polar distortion suggesting reflux nephropathy), renovascular abnormalities, and kidney masses and cysts.

p0265 Isolated microhematuria in a prospective donor necessitates consideration of thin basement membrane nephropathy, Alport syndrome (carrier status in women may cause minor or moderate abnormalities), and IgA nephropathy, as well as urinary tract infection, malignancy, and nephrolithiasis. Persistent microhematuria is relatively common and is evident in approximately 3% of the general population.⁴⁷ Among the possible disorders, IgA nephropathy is generally a contraindication to live donation, whereas thin basement membrane nephropathy may not necessarily be so.⁴⁷ The implications of isolated mesangial IgA without other manifestations of glomerulonephritis requires further research, and donation should be decided in the context of family history, absolute kidney function, presence of interstitial disease, and age. If persistent isolated asymptomatic microhematuria is detected during living donor evaluation, a workup should include cystoscopy and urinary cytology. A kidney biopsy also should be considered because glomerular hematuria cannot otherwise be excluded. If there is a possibility of familial disease (e.g., Alport syndrome, IgA nephropathy), this also helps clarify the prospective donor's future risk for progressive kidney disease.^{5,47}

p0270 Those with a history of bilateral or recurrent stones and those with systemic conditions associated with recurrent stone disease should not donate. An asymptomatic potential donor with a current single stone is suitable if the donor does not have a high risk for recurrence, if the stone is less than 1.5 mm, and especially if the stone is potentially removable during transplantation.⁴⁷ The evaluation of an asymptomatic donor with a single prior episode of nephrolithiasis should include evaluation of serum calcium, creatinine, albumin, and parathyroid hormone levels; spot urine for cystine; urinalysis and urine culture; spiral CT scan; chemical analysis of the stone, if available; and 24-hour urine measurement of oxalate, uric acid, and creatinine.

p0275 Atherosclerotic kidney vascular disease is a relative contraindication to living donation. If it is discovered, the donor should be normotensive, have normal kidney function, and have only unilateral disease. Careful evaluation for CAD and peripheral vascular disease should be undertaken, given the significant association of renovascular disease with atherosclerosis elsewhere. Fibromuscular dysplasia is found in 2% to 4% of prospective donors. Donors with severe and diffuse disease should not be accepted for donation. The age of the prospective donor should be considered, with the outcome in donors aged older than 50 years more benign than in younger donors.⁵¹

s0140 **Malignancy**

p0280 A history of certain malignancies is a contraindication to live kidney donation. These include melanoma, testicular cancer, kidney cell cancer, bronchial and breast cancer, choriocarcinoma, hematologic malignant neoplasm, and multiple myeloma.^{45–47} A history of malignancy may be acceptable for donation if prior treatment does not decrease kidney reserve, place the donor at increased risk for kidney disease, or increase the operative risk for nephrectomy. A history of malignancy may be acceptable if the specific cancer is curable and transmission of the cancer can be reasonably excluded; consultation with an oncologist may be required. Examples of cancers with low risk for transmission include certain mild forms of prostate, bladder, and cervical cancer. Consent to receive a kidney transplant must

include a discussion with the donor and the recipient indicating that risk for transmission of malignancy cannot be completely excluded.

Cardiovascular and Pulmonary Disease

In prospective donors, the cardiac assessment should be based on the history, risk factors, examination, and electrocardiographic findings.⁴⁷ An exercise or pharmacologic stress test and echocardiography may be warranted in certain circumstances. Individuals with myocardial dysfunction or coronary ischemia are at increased perioperative risk and should generally not donate. Pulmonary contraindications to donation include chronic lung diseases that significantly increase the anesthetic risk. If indicated by history and examination, pulmonary function testing, echocardiography, or sleep studies should be performed.⁴⁷ All donors should cease smoking for at least 8 to 12 weeks before surgery to minimize the risk for postoperative pneumonia.

COMPATIBILITY AND IMMUNOLOGIC CONSIDERATIONS

Blood Group Compatibility

Historically, transplantation across incompatible blood groups had been avoided because of the risk for hyperacute rejection mediated by preformed anti-A or anti-B antibodies to the carbohydrate blood group antigens, expressed by endothelial cells and by red blood cells. In the last two decades, ABO-incompatible transplantation has become more widespread, initially based on excellent outcomes described by Japanese centers.⁵² “Desensitizing” the recipient can avert hyperacute rejection. This involves removal of blood group antibodies by plasma exchange or immunoadsorption to achieve target titers. Preemptive splenectomy or rituximab administration (anti-CD20 monoclonal antibody) also is often used; however, the need for these measures is not clear. Rejection is predicted by high initial antibody titers and high rebound titers early after transplantation.⁵³ A further period of plasma exchange or immunoadsorption after transplantation is commonly instituted, whereas other centers determine the need for these therapies preemptively based on posttransplantation antibody titers. With use of this protocol, patient and graft survival was initially reported as similar to that of blood group-compatible transplantation for the short to medium term (up to 9 years).⁵² A recent systematic review, including data on 7098 patients who underwent ABO-incompatible transplant, reports an increase in mortality, graft loss, and infection at 5-years posttransplantation relative to recipients of ABO-compatible transplants.⁵⁴ Although the increase in absolute risk is small, these findings have prompted some to refer ABO-incompatible pairs to paired exchange programs (see www.bts.org.uk).

Human Leukocyte Antigen Compatibility

Tissue typing of recipient and donor determines their HLA match. HLA antigens are coded on chromosome 6, with half (one haplotype) inherited from each parent. The major histocompatibility class I HLA-A and HLA-B and class II HLA-DR antigens have routinely been determined because rejection responses are thought to most commonly stem from mismatches at these alleles. However, an increasing awareness of the importance of immune responses to other HLA antigens has resulted in centers routinely looking for the presence of antibodies to HLA-C, HLA-DQ, and HLA-DP. A six-antigen (HLA-A, HLA-B, and HLA-DR) match confers a graft survival advantage compared with zero antigen matches for both deceased and living donor transplantation of 10% at 10 years.⁵⁵ A further advance has been the description of the immunologically important antigenic subunits of HLA (epitopes) that rejection responses are directed against.⁵⁶ Each HLA antigen has multiple epitopes that can be compared between donor and recipient

to determine the degree of epitope matching, which also has implications for transplant outcomes.⁵⁷ In addition to determination of HLA compatibility, crossmatching and screening for anti-HLA antibodies are performed to assess the risk for rejection.

s0165 **Assessing Human Leukocyte Antigen Sensitization**

p0300 Sensitization to HLA antigens generally occurs through blood transfusion, pregnancy, or prior transplantation. The extent of HLA sensitization in an individual can be measured using complement-dependent cytotoxicity (CDC), enzyme-linked immunosorbent assays (ELISA), and antigen-coated bead technology; the latter is most common. These assays determine the number and specificity of antibodies against HLA antigens present in sera from the potential recipient and compare them to the HLA of a panel of individuals from the recipient's prospective donor population. A panel reactive antibody (PRA) of 75% means that the recipient has anti-HLA antibodies that react with 75% of people from that population. To avoid donors against whom they have donor-specific antibodies, patients with higher PRAs typically wait longer for deceased donor transplantation.

p0305 Presence in the recipient of antibodies to donor-specific HLA antigens can result in hyperacute rejection. Crossmatching of donor lymphocytes with recipient serum allows screening for this possibility. Terasaki and coworkers pioneered the CDC crossmatch,⁵⁸ which determines the presence of clinically significant antibodies by mixing donor T or B lymphocytes with recipient serum in the presence of complement. The sensitivity of the assay can be augmented by the addition of antihuman globulin. Presence of a positive T-cell CDC crossmatch to the donor is highly predictive of hyperacute rejection,⁵⁸ whereas a B-cell CDC crossmatch is more subject to false-positive results but should prompt a search for donor-specific antibodies (Fig. 107.4).^{59,60} A positive T-cell CDC crossmatch is an absolute contraindication to transplantation. The CDC crossmatch has limitations in detecting low level and non-complement-fixing antibodies, high false-positive rates, and logistical issues with obtaining reagents and running the assays in a timely fashion.

p0310 The flow crossmatch is more sensitive than the CDC in detecting antibody capable of binding to donor T or B lymphocytes. Binding

of antibody from donor serum is detected by flow cytometry after probing with a fluorescein-labeled anti-immunoglobulin antibody (Fig. 107.5).⁶⁰ The strength of a positive flow crossmatch can be estimated by the degree of lymphocyte fluorescence that is measured in channel shifts. An increasing number of channel shifts is associated with an increasing risk of rejection. The predictive value of a positive flow crossmatch for rejection is less than that of a CDC crossmatch because of its increased sensitivity, and it does not assess the ability of the antibody to fix complement. Some deceased donor allocation systems now employ flow crossmatching, in conjunction with virtual crossmatching, in place of CDC crossmatching. Flow crossmatching is commonly performed in a living donor transplant workup. A positive-flow crossmatch (with a negative CDC crossmatch) is not an absolute contraindication to proceeding; however, it may lead to alteration of the immunosuppressive regimen (e.g., use of a desensitization protocol) to decrease the risk or severity of antibody-mediated rejection.

Testing for anti-HLA antibodies in the recipient's serum is increasingly used as a means of predicting rejection. This virtual crossmatch compares the specificity of the antibodies identified with the prospective donor's HLA typing. Donor-specific anti-HLA antibodies correlate with worse graft survival even in the setting of a negative crossmatch.⁶¹ Although antibodies can be detected through a variety of techniques, including CDC panel reactive antibody testing and ELISA, the more sensitive antigen-coated bead technique is now almost exclusively employed. Multiple microscopic beads, each with a unique color that corresponds to a single HLA antigen that coats its surface, are mixed with recipient serum and probed with a fluorescein-labeled anti-immunoglobulin antibody (Fig. 107.6). Beads that bind antibody are identified as positive by fluorescence, and their HLA antigen specificity is detected by the unique coloring of the bead.⁶⁰ A positive virtual crossmatch is one in which a donor-specific antibody is detected. The strength of positivity of the virtual crossmatch can be estimated by the degree of fluorescence of the positive bead; however, this is not always accurate. The decision on whether to proceed with transplantation in the context of this information is complex. Recommendations for detection and characterization of clinically relevant antibodies in solid organ transplantation are summarized at www.bts.org.uk.

p0315

Complement-Dependent Cytotoxicity Crossmatch

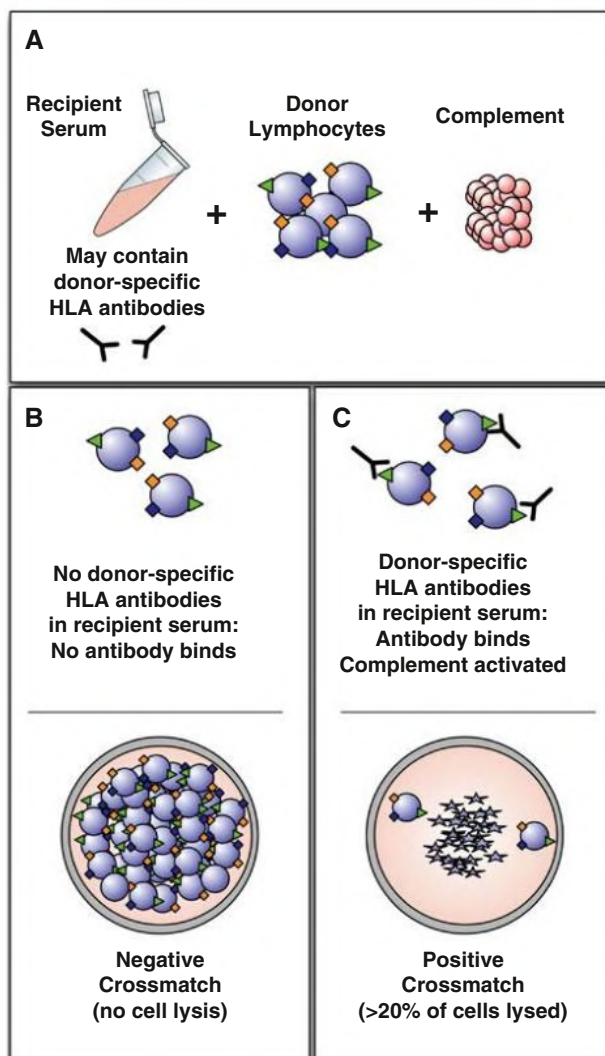


Fig. 107.4 Complement-Dependent Cytotoxicity (CDC) Crossmatch. (A) Recipient serum potentially containing donor-specific anti-human leukocyte antigen (HLA) antibodies is added to donor T or B lymphocytes, along with complement. (B) If donor-specific antibodies are not present, no lysis occurs and the result is deemed negative. (C) If donor-specific anti-HLA antibodies bind to the lymphocytes and then activate complement, cell lysis will occur and the crossmatch result will be deemed positive. The proportion of lysed cells is assessed, and the crossmatch is graded as being weakly, moderately, or strongly positive. (From Mulley WR, Kanellis J. Understanding crossmatch testing in organ transplantation: a case-based guide for the general nephrologist. *Nephrology*. 2011;16[2]:125–133.)

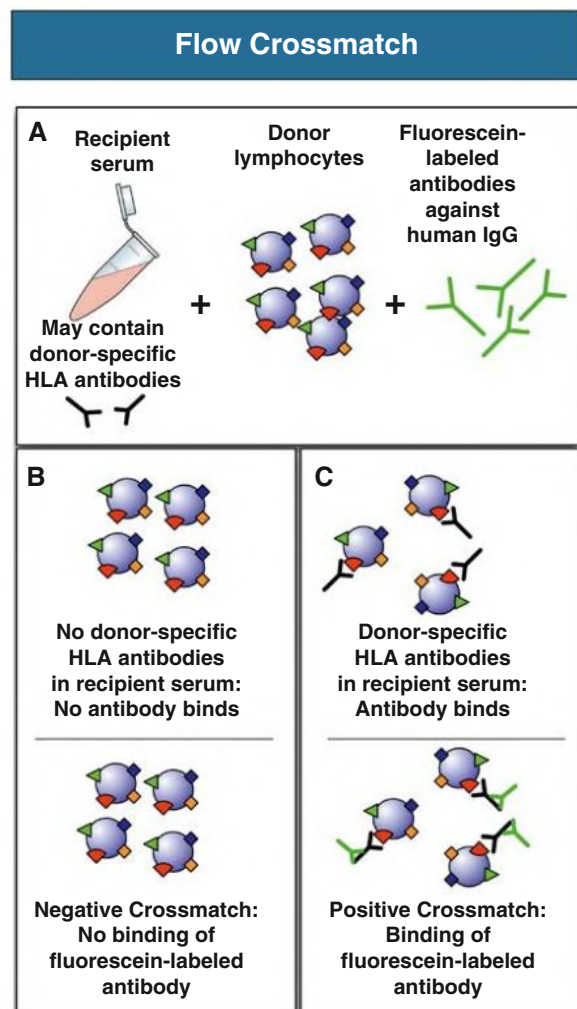


Fig. 107.5 Flow Crossmatch. (A) Recipient serum potentially containing donor-specific anti-human leukocyte antigen (HLA) antibodies is added to donor T or B lymphocytes, along with fluorescein-labeled antibodies against human immunoglobulin G (IgG). (B) If donor-specific antibodies are not present, no binding occurs and the result is deemed negative. (C) If donor-specific anti-HLA antibodies bind to the lymphocytes, these can then bind the fluorescein-labeled antihuman IgG antibody, and this will be detectable by flow cytometry. The strength of the fluorescence can be measured and expressed as “channel shifts” above the control sample. (From Mulley WR, Kanellis J. Understanding crossmatch testing in organ transplantation: a case-based guide for the general nephrologist. *Nephrology*. 2011;16(2):125–133.)

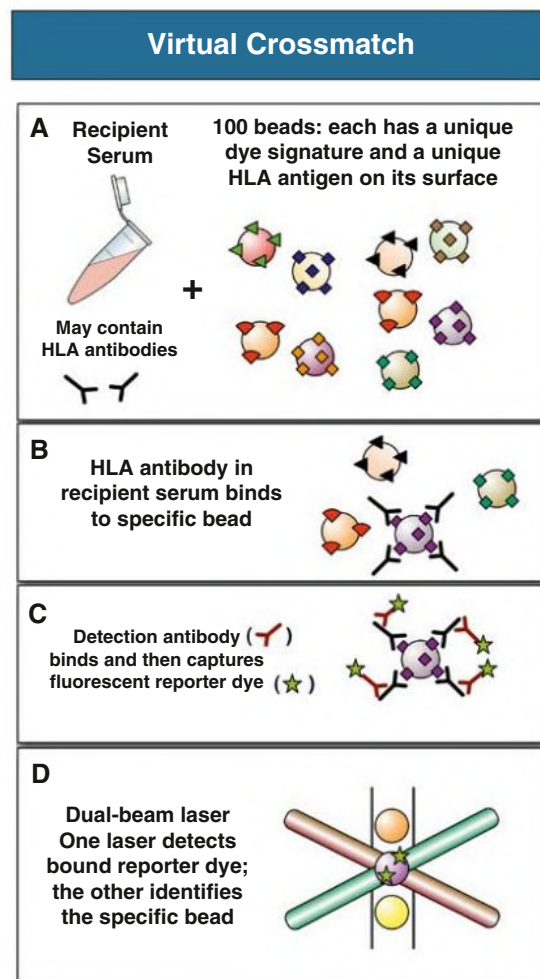


Fig. 107.6 Virtual Crossmatch. (A) Recipient serum potentially containing anti-human leukocyte antigen (HLA) antibodies is added to a mixture of synthetic beads. Each bead is coated with a set of antigens (screening beads) or, for more precise detail, with a single antigen (single antigen beads). A unique dye signature (up to 100) specifies the identity of each bead. (B) If anti-HLA antibodies are present, they will bind to the appropriate bead. (C) A detection antibody can subsequently bind and capture a reporter dye. (D) Each unique bead can then be interrogated for the presence of the reporter dye on its surface using a dual-beam laser. A profile of antibodies can thus be identified in the recipient and compared with the known HLA identity of any potential donor, allowing a prediction of the crossmatch result. (From Mulley WR, Kanellis J. Understanding crossmatch testing in organ transplantation: a case-based guide for the general nephrologist. *Nephrology*. 2011;16(2):125–133.)

SELF-ASSESSMENT QUESTIONS

- Which one of the following recipient factors is an absolute contraindication to transplantation?
 - Active sepsis
 - HIV infection
 - Obesity
 - Age older than 70 years
 - Previous malignancy

- In predicting the risk for cardiac death in potential kidney transplant recipients, stress echocardiography performs best in terms of which of the following parameters?
 - Sensitivity
 - Specificity
 - Positive predictive value
 - Negative predictive value
 - A and B equally

- o0070 3. Which one of the following brain-dead donors meets the definition for an expanded criteria deceased donor (as previously used in the United States)?
- o0075 A. A previously well 53-year-old person with a serum creatinine at donation of 1.7 mg/dL (150 μ mol/L)
- o0080 B. A 48-year-old person with diabetes, hypertension, vascular disease, and smoking history
- o0085 C. A 48-year-old person with hypertension and death from a cerebrovascular accident
- o0090 D. A 62-year-old person with no significant medical history
- o0095 E. A 58-year-old person with hepatitis
- o0100 4. Which one of the following investigations would you *not* routinely perform on a 67-year-old man to assess his suitability to donate a kidney to his wife?
- o0105 A. Urine microscopy
- o0110 B. Colonoscopy
- C. Hepatitis serology o0115
- D. Electrocardiogram o0120
- E. Kidney imaging o0125
5. Which one of the following will *best* identify a high risk for hyper-acute rejection? o0130
- A. Recipient tissue typing o0135
- B. Positive flow crossmatch o0140
- C. Positive T-cell cytotoxicity-dependent crossmatch o0145
- D. Positive virtual crossmatch o0150
- E. Donor O blood group o0155
-

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